

To: Eshwar

From: Aditya Alur

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Subject: Revised Research Proposal: Spectral Error Indicators for Neural PDE Solvers

Dear Eshwar,

Please find below the revised and complete research proposal for the project titled "Spectral Error Indicators for Neural PDE Solvers." This version includes a project abstract, explicit ML model framing, proposed architecture, validation datasets, and a relation to prior work.

Research Proposal

SPECTRAL ERROR INDICATORS FOR NEURAL PDE SOLVERS

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0. ABSTRACT

Physics-Informed Neural Networks (PINNs) can produce solutions that appear globally accurate while concealing large local errors — a failure mode that is difficult and expensive to detect without ground-truth references. This project proposes a spectral error indicator framework that analyzes intermediate PINN training snapshots using Dynamic Mode Decomposition (DMD) and Koopman operator theory. Specifically, we extract spectral features — modal energy distribution, eigenvalue drift, and spectral entropy — from sequences of solution snapshots across training epochs, and train a lightweight regression model to map these features to local PDE solution error. The result is a solver-agnostic, post-hoc error estimator that can flag high-error spatial regions and drive adaptive collocation point placement, without requiring re-solving the PDE or access to ground truth.

1. PROBLEM STATEMENT

PINNs and neural operators have emerged as powerful mesh-free alternatives to classical numerical PDE solvers. However, unlike Finite Difference or Finite Element methods, they currently lack reliable, cheap error certification mechanisms. A PINN solution may satisfy boundary conditions and look smooth globally while making large errors in localized regions — near sharp gradients, discontinuities, or undersampled collocation zones.

This project addresses that gap by developing a spectral error estimator that operates on DMD/Koopman decompositions of training snapshots, without requiring knowledge of the true solution. The central research question is: can changes in modal energy distribution, eigenvalue drift, or spectral entropy across PINN training snapshots serve as reliable indicators of local PDE solution error — and can these indicators guide adaptive mesh/collocation refinement?

2. ML MODEL FRAMING: INPUTS, OUTPUTS, AND PURPOSE

INPUTS:

- A sequence of K PINN solution snapshots recorded at regular training epochs (each snapshot is the network's predicted solution evaluated on a fixed spatial grid).
- DMD-derived features computed from this snapshot sequence:
 - **Spatial DMD modes** (dominant patterns across training)
 - **Eigenvalue trajectories** (growth/decay rates of each mode over epochs)
 - **Modal energy spectrum** (how energy is distributed across modes)
 - **Spectral entropy** (a scalar summary of energy concentration/dispersion)

OUTPUT:

- A spatial error map: a scalar field over the problem domain estimating the local absolute error $|u_{PINN}(x) - u_{true}(x)|$ at each point x .
- Optionally: a binary high-error region mask for adaptive refinement decisions.

PURPOSE OF THE MODEL:

During training, a PINN's solution landscape evolves. Regions where the solution is converging cleanly will show stable, low-entropy spectral behavior. Regions where the PINN is struggling — due to insufficient collocation points, sharp gradients, or conflicting loss

terms — will exhibit eigenvalue drift and rising spectral entropy. The model learns to recognize these spectral signatures and translate them into spatial error predictions. This enables:

1. Early stopping or targeted retraining before full convergence.
2. Adaptive collocation — adding points where the model predicts high error.
3. Uncertainty quantification without ensemble methods or Bayesian inference.

3. PROPOSED MODEL ARCHITECTURE

The framework consists of two sequential components:

COMPONENT A — Spectral Feature Extractor (DMD Pipeline)

- **Input:** Snapshot matrix $X \in \mathbb{R}^{N \times K}$, where N = spatial grid points, K = epochs
- **Process:** Standard DMD / Exact DMD decomposition
- **Output:** Per-mode eigenvalues λ_i , spatial modes ϕ_i , modal energies E_i , spectral entropy H computed over the eigenvalue spectrum
- **Note:** This component is unsupervised and requires no training.

COMPONENT B — Spatial Error Regression Head

- **Input:** Concatenated feature vector per spatial point x_j :
 $[\phi_1(x_j), \dots, \phi_r(x_j), E_1, \dots, E_r, H, |\lambda_1|, \dots, |\lambda_r|]$
 where r = number of retained DMD modes (rank truncation)
- **Architecture Options:**
 - *Option 1 (Baseline):* Gradient Boosted Regressor (XGBoost) — interpretable, fast to train, good for tabular spectral features.
 - *Option 2 (Spatial awareness):* Small CNN or U-Net applied to the 2D mode fields ϕ_i , producing a dense error map directly.
 - *Option 3 (Graph-based, future):* GNN if the collocation points are irregular / unstructured.
- **Output:** Error estimate $\hat{e}(x_j) \in \mathbb{R}^+$ for each spatial point.
- **Training:** Supervised — ground truth error $|u_{PINN}(x) - u_{analytical}(x)|$ computed on synthetic benchmark problems.

Adaptive Refinement Module (Downstream):

Thresholds $\hat{e}(x_j)$ to identify high-error regions → adds collocation points in those regions → PINN is retrained on updated collocation set. This loop runs until error estimates fall below a tolerance.

4. VALIDATION DATASETS

DATASET 1 — SYNTHETIC BENCHMARK (PRIMARY VALIDATION) ★

- **Problem:** 1D/2D Burgers' Equation with manufactured solution.
- **Why this:** Analytical solution is known exactly, so ground-truth error maps are computable at zero cost. We can deliberately construct cases where PINNs are known to struggle (sharp shock fronts, high Reynolds number) and verify that our spectral indicators correctly flag those regions. This is the dataset we build and trust first — if the model cannot work here, it cannot work anywhere. Full control over difficulty, noise, and geometry.
- **Source:** Self-generated using analytical solution: $u(x,t) = -2\nu (\partial/\partial x)[\ln \phi(x,t)]$ (Cole-Hopf transformation).
- **Variants:** Vary viscosity ν to control shock sharpness; vary collocation density to create intentional error hotspots.

DATASET 2 — ALLEN-CAHN / REACTION-DIFFUSION EQUATION

- **Problem:** Phase-field equation with stiff nonlinearity.
- **Why this:** Well-known PINN failure benchmark — the sharp interface is notoriously hard to resolve and produces large local errors. Widely used in PINN literature, enabling direct comparison.
- **Source:** Standard benchmark from Raissi et al. (2019) and subsequent PINN literature; data available at github.com/maziarraissi/PINNs.

DATASET 3 — NAVIER-STOKES (CYLINDER WAKE / LID-DRIVEN CAVITY)

- **Problem:** 2D incompressible Navier-Stokes.
- **Why this:** Tests generalization to a vector-valued, multi-physics PDE. Validates that spectral indicators remain meaningful in higher-dimensional, more complex solution landscapes.
- **Source:** CFD reference data from the FEniCS/OpenFOAM community benchmarks; also available via the DeepXDE library example suite.

5. RELATION TO PRIOR WORK

Prior Work	What They Do	Gap This Project Fills
Raissi et al. (2019) <i>PINNs, J. Comp. Physics</i>	Original PINN formulation — physics loss + data loss	No error estimation or adaptive refinement
Dwivedi & Srinivasan (2020) <i>Res.-based adaptive PINN</i>	Adaptive collocation via PDE residual magnitude	Uses residual directly — residual $\neq$ true error in stiff/nonlinear problems
Lu et al. (2021) <i>DeepXDE library</i>	DeepXDE: residual-based adaptive point selection	Same limitation: residual-driven, not spectrally informed
Schmid (2010) <i>DMD, J. Fluid Mechanics</i>	Original DMD formulation for fluid dynamics	Applied to fluid snapshots, not to NN training dynamics
Kutz et al. (2016) <i>Dynamic Mode Decomposition</i>	DMD book — Koopman theory and data-driven dynamics	No connection to neural network error estimation
Yang & Perdikaris (2021) <i>Bayesian PINNs</i>	B-PINNs: Bayesian PINNs for uncertainty quantif.	Expensive (requires ensemble/MCMC); not solver-agnostic
THIS PROJECT	DMD on training snapshots → spectral features → spatial error regression	Lightweight, solver-agnostic, no ground truth needed at inference time

6. EXPECTED OUTCOMES

- Validated correlation between DMD spectral features and local PINN error on synthetic Burgers' benchmarks.

- A trained spatial error regression model with quantified performance (MAE, spatial precision/recall of high-error region detection).
- An adaptive refinement loop demonstrating accuracy improvement over uniform collocation baselines.
- Benchmarking results across all three datasets.
- Technical report; draft manuscript if results are publication-ready.

7. RISKS AND OPEN QUESTIONS

- DMD on training-epoch snapshots is non-standard; the method was designed for temporal dynamics, not optimization trajectories. Early experiments will validate whether meaningful spectral structure exists in this setting.
- Ground-truth error is only available for problems with analytical solutions; the framework's inference-time use (no ground truth) will be validated separately by holding out the true solution during error prediction.
- Solver-agnostic claims will be scoped to PINN architectures initially before extension to neural operators (FNO, DeepONet).

I look forward to your feedback and am happy to discuss any aspect of this proposal at your convenience.

Warm regards,

Aditya Alur

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